MultiQT notes

November 4, 2026

1 Studentized score

Writing the score as a signed residual gives the usual additive MultiQT correction. Studentizing that residual lead to multiplicatively scaled offsets in a MulitQT update or offsets in a scaled coordinate system. With changing scale, the accumulated correction as well as the new offset are scaled. Thus, a positive offset that was built up in a underprediction/ upswing phase can easily be removed after the peak (so the idea). With quantile specific scales the ordering constraint lead to the feasible set being a diagonally transformed shifted isotonic cone. This is still an isotnic cone and a closed convex polyhedron with a unique Euclidean projection (weighted isotonic regression). The gradient steps now optimize weighted quantile loss. The coverage criterion and calibration target are unchanged (strictly increasing invertible transformation).

2 question and ides

- What other scores would be reasonable to consider, i.e. what property of the base forecast do I need to recalibrate? Instead of asking for 1d PIT calibration, could I ask for 1d hdr-calibration? With studentized scores, or in general strictly increasing and invertible scores, the coverage indicator and calibration target do not change